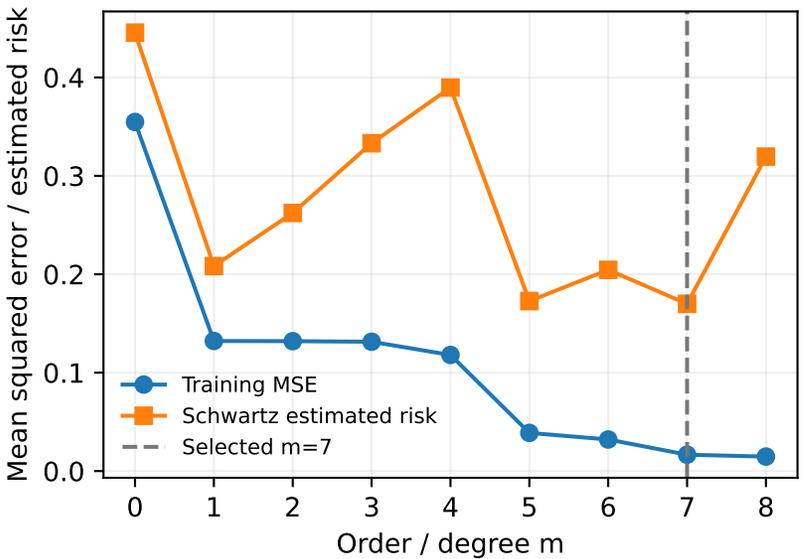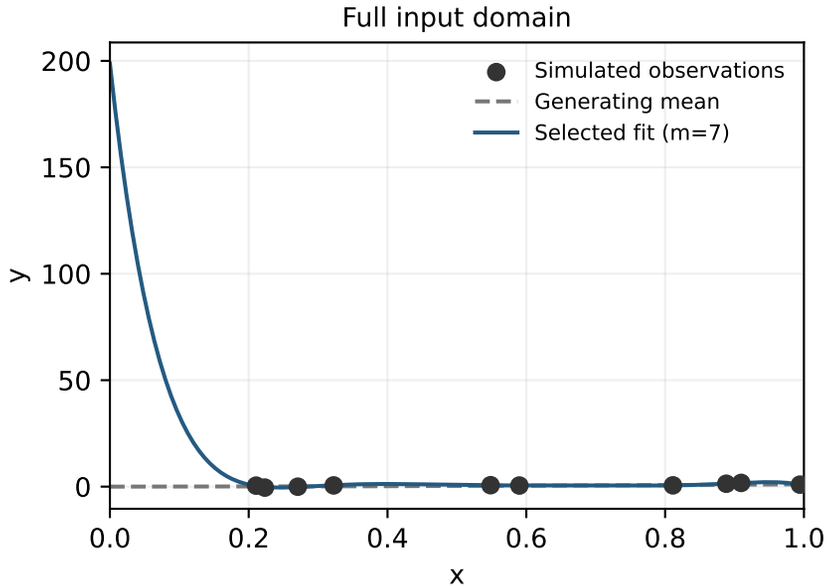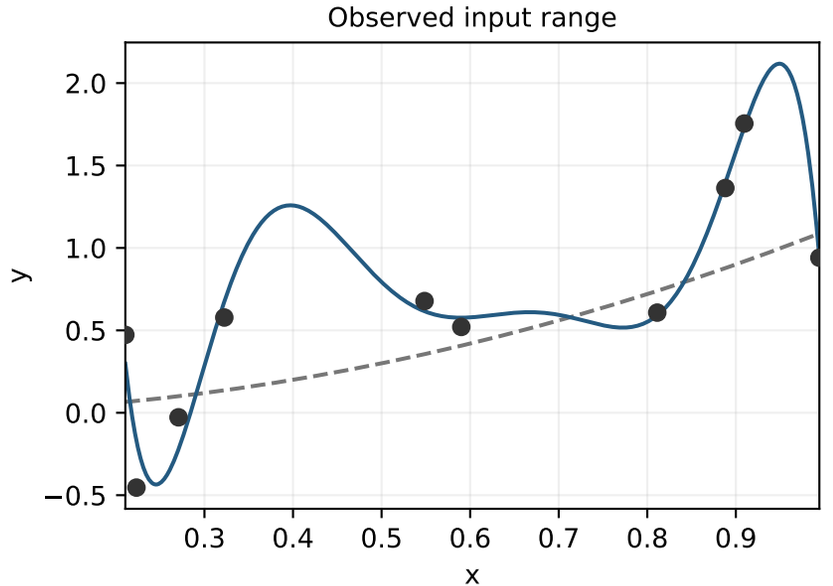Problem 2.11: algebraic models (simulated realization)